

What Can These Data Support?

AP Statistics · Unit 1 · Topic 1.1 · TEACHER KEY

CED TETHER

- **VAR-1.A** Identify questions to be answered, based on variation in one-variable data. [Skill 1.A]
- **VAR-1.A.1** Numbers may convey meaningful information, when placed in context.

Essential question: How can leaving out valid data change the answer to a question?

DOK-3 skill: 4.A · **FRQ pattern:** claim-vs-data-support-bound-the-conclusion

DOK rationale: Part (c) weighs two claims against the full data and a contextual threshold, and explains how excluding valid observations changes the conclusion.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Who is included in each percent?
- Why does the total change when two observations are removed?
- Were the removed values recording errors?
- Which conclusion answers the original question about all 12 students?

[WARN] LOOK FOR

- Dividing by 12 after removing two students
- Calling the shortened percent an arithmetic error rather than a changed data set
- Treating a valid low value as permission to omit it

SCORING PART (c) — E / P / I

Expected elements

- **applies-threshold** (required) — Supports Claim 1 using 25 percent versus the 20 percent rule
- **interprets-shortened-data** (required) — Recognizes 10 percent describes the shortened table, not all 12 students
- **explains-omission** (required) — Explains how removing valid low-sleep observations hides the original concern

E	All three elements are correct and linked to this problem. Accept equivalent plain-language reasoning.
P	At least one element includes a correct evidence-to-reason link, but another is missing or incorrect.
I	No correct evidence-to-reason link; only a choice, copied numbers, or unsupported statements.

Common mistakes

- Dividing by 12 after removing two students

- Calling the shortened percent an arithmetic error rather than a changed data set
- Treating a valid low value as permission to omit it

[STAR] THE PROBLEM — annotated

Suppose a school newspaper records last night's sleep for 12 students in one class. All 12 values are valid. For this class, the school flags a sleep concern if more than 20% slept fewer than 6 hours.

Claim 1: "The full table shows a sleep concern in this class."

Claim 2: "Leave out the two lowest sleep values and there is no sleep concern."

Sleep last night

Student	Sleep (hours)
Aiden	5.0
Bea	7.0
Cruz	8.5
Dev	6.5
Elise	8.0
Fiona	7.5
Gabe	5.5
Hana	9.0
Imani	7.0
Jalen	6.0
Kira	8.0
Leo	4.5

FIRST TAKE — one sentence before you work the parts

Which claim, if either, would you print? Name one detail behind your choice.

Key: Not graded. Students can choose using the sleep values and the printed threshold before formal vocabulary.

[BOOK] USE DATA IN CONTEXT (keep on the page)

A variable is a characteristic recorded for each individual. Variation means the values differ. A threshold is a cutoff used to decide whether action is needed. Percent = $100 \times \frac{\text{count meeting the condition}}{\text{total count}}$. When observations are removed, count the remaining observations and use the remaining total. Interpret the result in the original context.

DOK 1 (a) Name the variable and its unit.

Key: Hours of sleep last night for each student in this class.

DOK 2 (b) Find the percent who slept fewer than 6 hours using all 12 students. Then leave out the two lowest values and find the new percent.

Key: All data: $3/12 = 25\%$. Removing 4.5 and 5.0 hours leaves $1/10 = 10\%$ below 6 hours.

DOK 3 (c) Which claim would you print? Use both percents and the school's rule. Explain why keeping or removing the two values changes what readers learn about this class.

Key: Print Claim 1: 25% exceeds the 20% threshold. Claim 2's 10% is arithmetically correct for the shortened table, but excluding two valid low values changes both the count and the total. It hides part of the sleep concern in the original class.